

IPS → Rapid Threat Containment

C6 — a custom Snort 3 intrusion rule auto-quarantines a live SD-Access endpoint via ISE ANC → RADIUS CoA · detection-driven, no human in the loop

RUN DATE	TESTER	DESIGN	PLAN
2026-07-17	testing-agent (live)	SD-Access ISE Integration	RTC-020...025 (Stage C)

VERDICT PASS

6 PASS · 0 partial · 0 FAIL. A live 802.1X endpoint (**alice**) was auto-contained in seconds when the FTD's **Snort 3 engine dropped a payload signature** (SID 1000001) — the intrusion event drove an FMC correlation rule whose remediation **auto-applied an ISE ANC quarantine over pxGrid**, firing a **RADIUS CoA** that terminated her fabric session (0% → 100% loss). She was then cleanly released and fully restored. Read-only on built config; only reversible ANC round-trips, all undone.

1. Executive summary

Live, independent re-verification of roadmap item **C6 — IPS / Snort 3 → Rapid Threat Containment** (Stage C of the rapid-threat-containment plan, RTC-020...025) on the **SD-Access ISE Integration** lab. A custom **Snort 3 intrusion rule** (SID 1000001) on the FTD, matched by live host traffic, auto-quarantines the offending endpoint through the existing RTC loop — an **IPS *detection*** (not just an ACL deny) driving the containment. The subject is a single live, already-authenticated 802.1X endpoint — **alice / HOST1** (172.16.10.50 , MAC 52:54:00:03:0B:0D) on fabric edge **EDGE1**.

The chain, observed end to end. Firing `printf 'C6TRIGGER probe' | nc -w 3 198.18.128.51 8000` from HOST1 made the FTD Snort 3 engine **drop** the packet (Blocked-Packets 7 → 9) and raise an **intrusion event**. With **no human in the loop**, the FMC correlation rule `Quarantine-on-IPS-C6` matched **Rule SID 1000001** and its remediation `Quarantine-Source` **auto-applied an ISE ANC Quarantine over pxGrid** on alice's MAC. ISE fired a **RADIUS CoA (RFC 5176)** — MnT event **5205**, `Error-Cause=200` — that **terminated her session** (active sessions 1 → 0, EDGE1 port with no session, HOST1 → host-splunk 0% → 100% loss).

Result: 6 PASS · 0 partial · 0 FAIL. Clearing the ANC and reassociating the supplicant fully restored alice to Employees / SGT 4 (session back, 0% loss, no ANC residual). **alice is verified restored.** No built configuration was modified — read-only plus reversible ANC round-trips, all undone.

2. Scope & systems under test

Test plan: Lab Designs / rapid-threat-containment — **new Stage C** (RTC-020...025), extending v2.0 (Stage A RTC-001...007 , Stage B RTC-010...014). **Design:** C6 on the SD-Access ISE Integration lab; context module `Custom Designs/SD-Access ISE Integration/modules/fmc-ips-rtc.md`.

What the lab is (plain English). A virtual **Cisco SD-Access campus fabric** (a LISP/VXLAN overlay with a control-plane/border node **BORDER-CP** and a fabric edge **EDGE1**, fused to the outside by **FUSION-R1**) with a full identity + security stack: **Cisco ISE 3.5** as the RADIUS policy server, TrustSec and pxGrid controller; **Microsoft AD** (`mitchcloud.lab`) as the external identity store; a **Cisco Secure Firewall (FMC + FTD)** at the fabric's external edge running **Snort 3** intrusion prevention; and **Splunk** as the telemetry sink. Employee endpoint `alice` authenticates with **802.1X (PEAP)**, is placed in the **Employees** security group (SGT 4) under **Closed Authentication**, and gets normal fabric access.

What C6 adds. C2 proved the ANC → CoA containment primitive, triggered by an operator (Stage A) or an FMC correlation on a **connection/ACL** event (Stage B). **C6** adds a **detection-based** trigger into the *same* RTC-Quarantine correlation policy: a custom Snort 3 rule that fires on a payload signature, so an **IPS drop** now auto-quarantines the source.

```
HOST1 (alice 172.16.10.50) sends "C6TRIGGER"
→ FTD Snort custom rule SID 1000001 → DROP + intrusion event
→ FMC correlation rule "Quarantine-on-IPS-C6" (condition: Rule SID is 1000001)
→ response "Quarantine-Source" (ANC Policy for Source, pxGrid EPS)
→ ISE ANC Quarantine on alice's MAC → CoA → 802.1X session terminated → 100% loss
```

Component versions verified live this run

Component	Version
Cisco ISE (PSN / pxGrid)	3.5.0.527
Cisco Secure Firewall Management Center (FMC)	10.0.1 (build 1) · vdb 433, sru 2026-07-15
Cisco Secure Firewall Threat Defense (FTD)	10.0.0 — Snort 3, PREVENTION, base <i>Connectivity Over Security</i>
Fabric edge / NAD (EDGE1)	cat9000v IOS-XE 17.18
SD-Access fabric	LISP/VXLAN, Closed Authentication

3. Test environment

Lab: CML “SDA-Fabric” (77dd2fde-1fda-4cc9-9b29-48ff98bd1395) — STARTED / converged, all nodes BOOTED. IPs are **lab-specific**.

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / VLAN
HOST1 (alice)	802.1X test subject · alpine	—	172.16.10.50 · MAC 52:54:00:03:0B:0D	CAMPUS_VN (VLAN 1021)
EDGE1	fabric edge / NAD + CoA client · cat9000v	198.18.128.73	RADIUS/CoA id 10.1.0.3 · port Gi1/0/3	CAMPUS_VN

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / VLAN
BORDER-CP	fabric control-plane + border · cat9000v	198.18.128.72	LISP map-server / L3 handoff	CAMPUS_VN / IOT_VN
FUSION-R1	fusion router (PBR to FTD) · cat8000v	198.18.128.71	inside 10.1.245.1/30	global
FTDv	Secure Firewall TD — Snort 3 IPS (SDA-IPS) · ftdv	198.18.128.81	inside → FUSION, outside → /18	routed
FMCv	Mgmt Center — correlation + remediation · fmcv	198.18.128.80	—	—
ISE 3.5 PSN (ise35)	policy + CoA origin (UDP 1700) + pxGrid	198.18.134.35	—	external VM
host-splunk	trigger dest (allow → Snort-inspected) · splunk	198.18.128.51	—	—

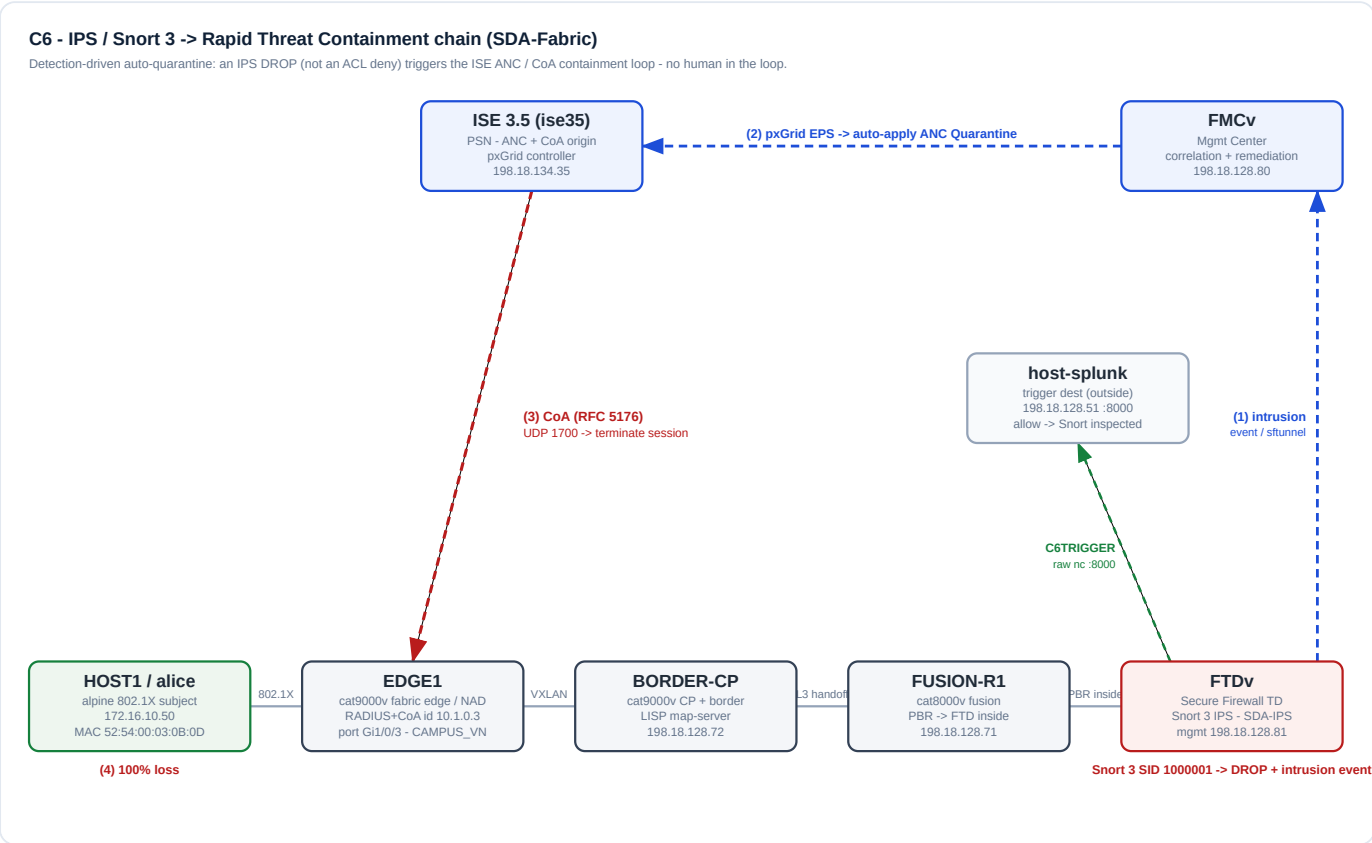


Figure 1 — C6 IPS-RTC chain. Green = the C6TRIGGER data flow the FTD inspects; red = the CoA that severs alice's session; blue = the FTD →FMC intrusion event and the FMC → ISE pxGrid auto-apply of the ANC quarantine. Auto-generated from the live lab topology + addressing.

4. Results — automated gate (MCP servers)

Not applicable to this run. C6 acceptance is **manual-live** end-to-end against the live ISE 3.5 + FMC/FTD + SDA fabric; there is no offline CI gate for a lab-design case. Underlying MCP tool health (ise-mcp, firepower-mcp, cml-mcp) is covered by those servers' own plans and the 2026-07-15 automated-gate report. All live surfaces answered this run: `ise_check_surfaces` → `openapi/mnt/ers` reachable; FMC token auth + config API OK; FTDv console + EDGE1/HOST1 pyATS consoles OK.

5. Results — lab-design acceptance

Manual-live, case-by-case. Trigger: `printf 'C6TRIGGER probe' | nc -w 3 198.18.128.51 8000` (raw payload stays in `pkt_data` ; fired x2 this run).

ID	Objective	Result	Evidence
RTC-020	Custom Snort 3 rule exists in FMC (Local Rules, gid 2000 / sid 1000001)	PASS	<code>GET object/intrusionrules/...972036</code> → name 2000:1000001 , gid 2000 , sid 1000001 , rev 5 , group Local Rules , <code>content: "C6TRIGGER"</code> , <code>nocase</code> , msg "LOCAL C6 RTC test trigger". Per-policy override in SDA-IPS = DROP .
RTC-021	IPS policy SDA-IPS attached to the allow rule (Snort inspects the flow)	PASS	FMC: <code>Permit-CAMPUS-Services action=ALLOW, ipsPolicy=SDA-IPS</code> , enabled. SDA-IPS = PREVENTION , base Connectivity Over Security (active base → compiles Local Rules). On-box <code>show access-control-config</code> : Rule: <code>Permit-CAMPUS-Services</code> · Action: <code>Allow</code> · Intrusion Policy: <code>SDA-IPS</code> (Source <code>net-campus10</code> → Dest <code>host-splunk 198.18.128.51</code>).
RTC-022	Baseline (uncontained): 1 active session, no ANC, HOST1 → host-splunk 0% loss	PASS	<code>active_session_count=1</code> ; <code>list_anc_endpoints=[]</code> ; HOST1 <code>eth0 172.16.10.50/24</code> ; ping 198.18.128.51 0% loss (4/4) ; ISE: <code>alice</code> active, Employees / SGT 4 , <code>PermitAccess</code> , rule <code>Employees_SGT</code> , PEAP(EAP-MSCHAPv2), store <code>mitchcloud</code> , audit-session <code>0217010A000000146FFB1A92</code> .
RTC-023	Fire C6TRIGGER → FTD Snort drops it (Blocked Packets increments)	PASS	<code>show snort statistics</code> Blocked Packets 7 → 9 (+2, one drop per trigger connection). Intrusion event raised (drives the RTC loop).
RTC-024	IPS drop auto-quarantines <code>alice</code> via the RTC loop (ANC + CoA + containment) — no human	PASS	No <code>ise_apply_anc</code> called. <code>list_anc_endpoints</code> [] → 2 records , both 52:54:00:03:0B:0D / Quarantine (auto-created). CoA: MnT msg 5205 "Dynamic Authorization succeeded" , <code>Error-Cause=200</code> , 12:37:25.647Z, audit-session <code>0217010A000000146FFB1A92</code> . <code>active_session_count=0</code> ; EDGE1 <code>Gi1/0/3</code> → <i>No sessions match</i> ; HOST1 → 198.18.128.51 100% loss .
RTC-025	Reversibility: clear ANC + reassociate → session restored, no residual, 0% loss	PASS	<code>ise_clear_anc(..., Quarantine)</code> x2 → <code>list_anc_endpoints=[]</code> . HOST1 <code>sudo -n wpa_cli -i eth0 reassociate</code> → EAP SUCCESS , Authorized. EDGE1 <code>Gi1/0/3</code> Authorized, alice, SGT 4, 172.16.10.50 , new session <code>0217010A000000157016CAEF</code> ; <code>active_session_count=1</code> ; HOST1 → 198.18.128.51 0% loss (5/5) after ~90 s LISP EID re-registration. No ANC residual.

6. Summary statistics

Metric	Value
Stage C cases executed	6
PASS	6
Partial / FAIL / Skip / Unreachable	0 / 0 / 0 / 0
Snort Blocked-Packets delta on trigger	+2 (7 → 9)
CoA outcome	MnT 5205 <code>Error-Cause=200</code> , session terminated (1 → 0)
Reachability flip (HOST1 → host-splunk)	0% → 100% → 0% loss
alice restored	SGT 4, 1 session, 0% loss, no ANC residual
Built config modified	None (read-only + reversible ANC round-trips, all undone)

7. Observations & defects

Defects raised: none. All six Stage-C cases passed and the endpoint was fully restored. Observations (operational notes, none downgrade a case):

- **Two ANC records auto-created for one MAC.** Firing the trigger twice produced two correlation fires and **two duplicate `ancendpoint` records** for `52:54:00:03:0B:0D` (ISE ERS permits duplicates per MAC), so full release needed **two `ise_clear_anc` calls**. A single trigger creates one record. Containment is identical — a cleanup nuance, not a defect.
- **HTTP-buffer nuance on port 8000.** The trigger's first data segment drew an HTTP 400 from Splunkd, yet the Snort Blocked-Packets counter still incremented and the intrusion event fired the RTC loop (gotcha below). The definitive outcome (auto-quarantine + 100% loss) is unambiguous; the raw `nc` payload is the reliable trigger.
- **LISP reconvergence ~90 s this run.** After release the control plane restored within seconds (EDGE1 Authorized / SGT 4, ISE live session) but the **data path** returned only after a further LISP EID re-registration cycle (~90 s vs the prior ~15–30 s). Timing caveat, not a defect.
- **FTDv health “yellow” (cosmetic).** Reported `health=yellow` but MANAGED/DEPLOYED and fully functional — it dropped the trigger and raised the event.

C6 gotchas (carried from the module, re-confirmed this run)

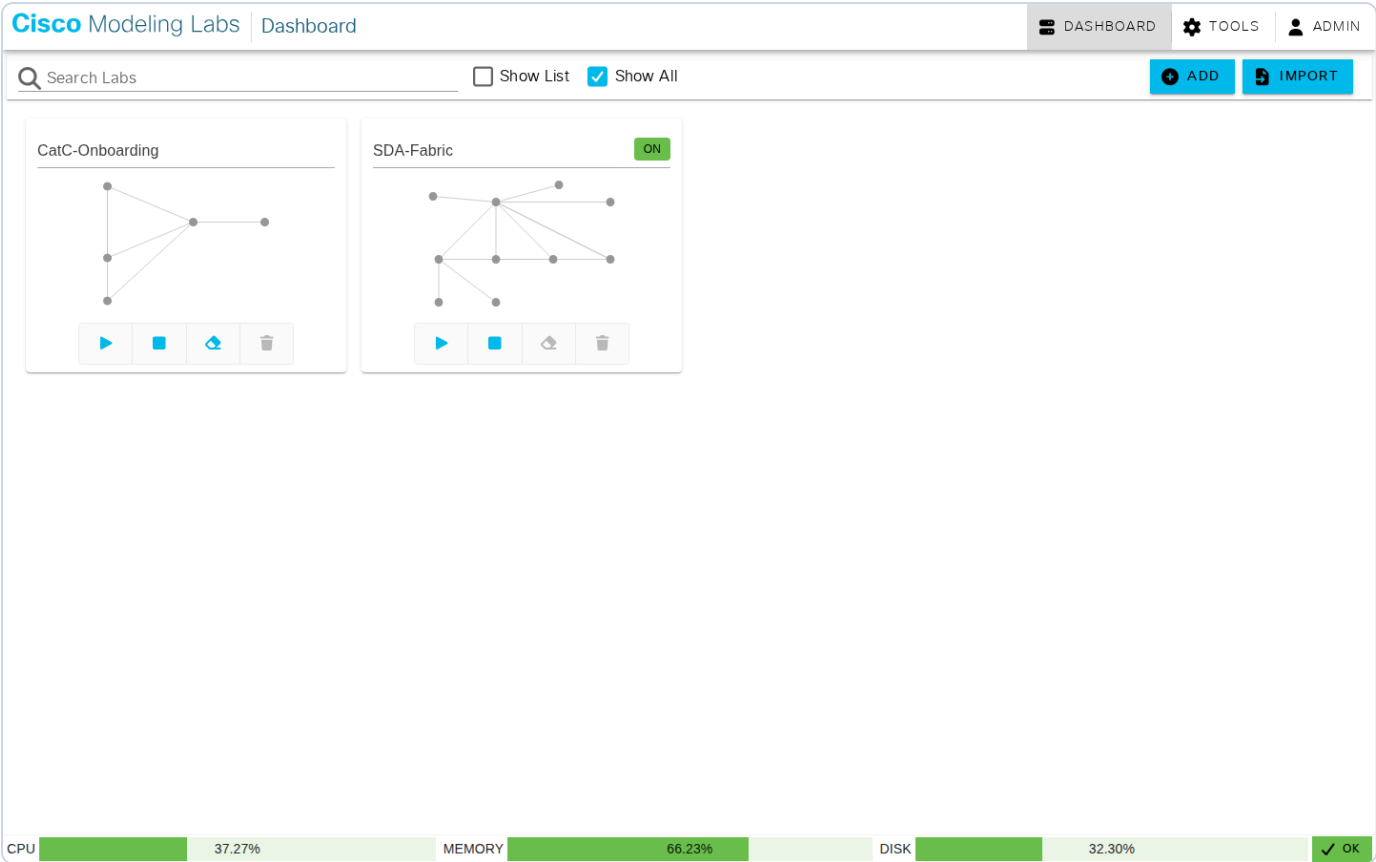
“No Rules Active” base does NOT compile custom Local Rules. SDA-IPS uses an **active** base (*Connectivity Over Security*) so the rule group compiles and the per-rule **DROP** override takes effect — otherwise Blocked Packets stays 0.

- **HTTP-inspected flows hide the payload from a bare `content`** — use a raw non-HTTP `nc` payload (bytes stay in `pkt_data`) or the `http_uri` sticky buffer.
- **Correlate on Rule SID, not Generator ID** — FMC files user rules under **GID 2000**, absent from the correlation Generator-ID picker; key on Rule SID is 1000001.

Remediation briefs: none required — no FAIL/gap to hand back.

8. Appendix

CML canvas capture (ground truth). The SDA-Fabric lab, STARTED and converged at run time.



Appendix A — CML dashboard showing the “SDA-Fabric” lab ON (converged). Ground truth for the topology in Figure 1.

Key raw evidence

```
FTDv# show snort statistics (before / after trigger)
Blocked Packets      7  ->  9

ISE MnT AuthStatus (52:54:00:03:0B:0D):
msg 5205 "Dynamic Authorization succeeded" {Error-Cause=200} 12:37:25.647Z
audit-session 0217010A0000000146FFB1A92 (alice's live session)

ISE ancendpoint (auto-created, no ise_apply_anc):
deldea5c-... -> 52:54:00:03:0B:0D / Quarantine
8434eee2-... -> 52:54:00:03:0B:0D / Quarantine

HOST1 -> 198.18.128.51 : 0% loss -> 100% loss -> 0% loss (restored)
EDGE1 Gi1/0/3 : Authorized(SGT 4) -> No sessions -> Authorized(SGT 4)
active_session_count : 1 -> 0 -> 1
```

results-C6.json holds the machine-collected raw results for all six cases. FMC REST GETs, FTDv show access-control-config / show snort statistics, ISE list_anc_endpoints / auth_status_by_mac / session lookups, EDGE1 show access-session, and HOST1 pings + wpa_cli were captured inline in §5.